

From Scratch Implementation of XGBoost & KNN

Coding Club IITG ML Module

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Motivation & Objective

- Understand ML algorithms by coding them from scratch
- Learn optimization and fundamentals of ML
- Goal: Implement KNN and XGBoost without using ML libraries

KNN Algorithm (From Scratch)

- Idea: Classify a point based on its k nearest neighbors
- Steps:
 - Compute distances (Euclidean/others)
 - Sort & pick k nearest
 - Majority vote / averaging
- Challenges: efficiency on large datasets, tie-breaking

XGBoost Algorithm (From Scratch)

- Idea: Boosting with decision trees using gradients and hessians.
- Steps:
 - Base learner = decision tree
 - Fit new trees on residuals (gradients)
 - Combine predictions with weights
- Challenges: gradients & hessians calculation, overfitting, runtime
- Implemented in C++ (not optimized yet)

Results & Learnings

- Tested on small datasets (e.g., Iris, Fish)
- Compared with sklearn for validation
- Key Learnings:
 - Inner working of ML models
 - Importance of optimization